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Universal light-sheet generation with field synthesis

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We introduce field synthesis, a theorem and method that can be used to synthesize any scanned or dithered light sheet, including those used in lattice light-sheet microscopy (LLSM), from an incoherent superposition of one-dimensional intensity distributions. Compared to LLSM, this user-friendly and modular approach offers a simplified optical design, higher light throughput and simultaneous multicolor illumination. Further, field synthesis achieves lower rates of photobleaching than light sheets generated by lateral beam scanning.

Confocal microscopy is widely used for subcellular volumetric imaging of biological specimens[1]. However, the exceedingly small dwell times of modern confocal microscopes (for example, <2 µs) necessitate high peak illumination intensities, which accelerate photobleaching and phototoxicity[2,3]. Further, regions above and below the laser focus are unnecessarily illuminated, resulting in the expenditure of precious photons that do not productively contribute to image formation. In contrast, light-sheet fluorescence microscopy (LSFM) adopts an orthogonal imaging geometry whereby the sample is illuminated from the side and fluorescence is collected at 90° with high-numerical-aperture optics and million-fold camera-based parallelization[4]. Because fluorescence excitation above and below the depth of focus is minimized, optical sectioning is automatically achieved, and unnecessary illumination of the specimen is avoided. Consequently, LSFM permits unparalleled long-term imaging of dynamic biological specimens.

In LSFM, critical tradeoffs exist among axial resolution (influenced by the thickness of the light sheet), field of view (that is, the length over which a beam can approximate a sheet), illumination duty cycle and illumination confinement. In an effort to simultaneously optimize these optical parameters, the Betzig laboratory developed LLSM[5]. Here, the specimen is coherently illuminated with multiple plane waves that sculpt a propagation-invariant two-dimensional (2D) optical lattice. In its most widely adopted operating mode, the optical lattice is dithered (that is, scanned over a small lateral distance) to form a time-averaged sheet of light. The alternative structured illumination microscopy mode of LLSM is seldom used (see Supplementary Table 1) because the moderate gain in resolution rarely outweighs the decreased acquisition rate. Unfortunately, the generation of optical lattices requires an optical train that is complex, expensive and lossy (~0.1–1% input laser power delivered to the sample)[5].

In this paper, we show that any scanned light sheet (that is, a time-averaged light sheet obtained by rapid scanning of a 2D intensity distribution laterally[6]), including dithered lattice light sheets, can be synthesized via a simple and universal optical process described by a mathematical theorem that we refer to as field synthesis (Supplementary Note 1). Applied to Bessel beam and lattice light sheets, field synthesis illuminates the entire field of view simultaneously during the exposure of the camera and results in less photobleaching than time-averaged light sheets generated by lateral beam scanning. When field synthesis is applied to the dithered mode of LLSM, light losses in the excitation train are reduced 15-fold (Supplementary Note 2) and simultaneous multicolor imaging becomes possible. Furthermore, the number of optical components is halved. No spatial light modulator, diffraction optics or polarization optics are needed, which substantially reduces the complexity and cost of an LLSM system (Supplementary Fig. 1 and Supplementary Note 2). We mathematically prove the field synthesis theorem, demonstrate proof-of-concept imaging and provide software and documentation so that the biological community may adopt the technology.

The essence of the field synthesis theorem is that the intensity of a scanned light sheet can be replicated by an averaged line scan in the Fourier domain. In the ideal line scan case, the projection of the intensity, the squared modulus, of a 2D function $F(x, z)$, representing the electric field at the front focal plane, as written on the right-hand side of the equation below, is equivalent to the average of the squared modulus of a line scan in the Fourier domain of the same function, $\hat{F}(k_x, k_z)$, as written on the left-hand side of the equation (Supplementary Notes 1 and 3 and Supplementary Videos 1 and 2):

$$\int_{-\infty}^{+\infty} \left|\mathcal{F}^{-1}\{\hat{F}(k_x,k_z)\,\delta(k_x-a)\}(x,z)\right|^2 \mathrm{d}a = \int_{-\infty}^{+\infty} \left|F(x',z)\right|^2 \mathrm{d}x'$$

where $\mathcal{F}^{-1}$ is the inverse Fourier transform, k_x and k_z are reciprocal coordinates, and (x, z) are real space coordinates. Here, the line scan $\delta(k_x - a)$, where a indicates the line position along k_x, follows the same direction in Fourier space, k_x, as the projection in real space, x. This has practical applications for scanned light sheets, where a time-averaged light sheet is obtained by lateral scanning of the illumination beam. In a conventional LSFM, the illumination beam is shaped by a light distribution with spatially varying amplitude and phase in the back focal (pupil) plane of the excitation objective, which can be described as 2D pupil function (Fig. 1a, top row)[7]. Our theorem predicts that if a focused line is scanned over the same pupil function, an identical time-averaged light sheet is obtained (Fig. 1a, bottom row). We also provide a generalized theorem describing non-ideal line scans with an arbitrary line profile (Supplementary Note 1). Thus, by scanning a focused line over an appropriate pupil mask (for example, a two-dimensional spatial filter), we can create any type of scanned or dithered light sheet, including those used by LLSM.

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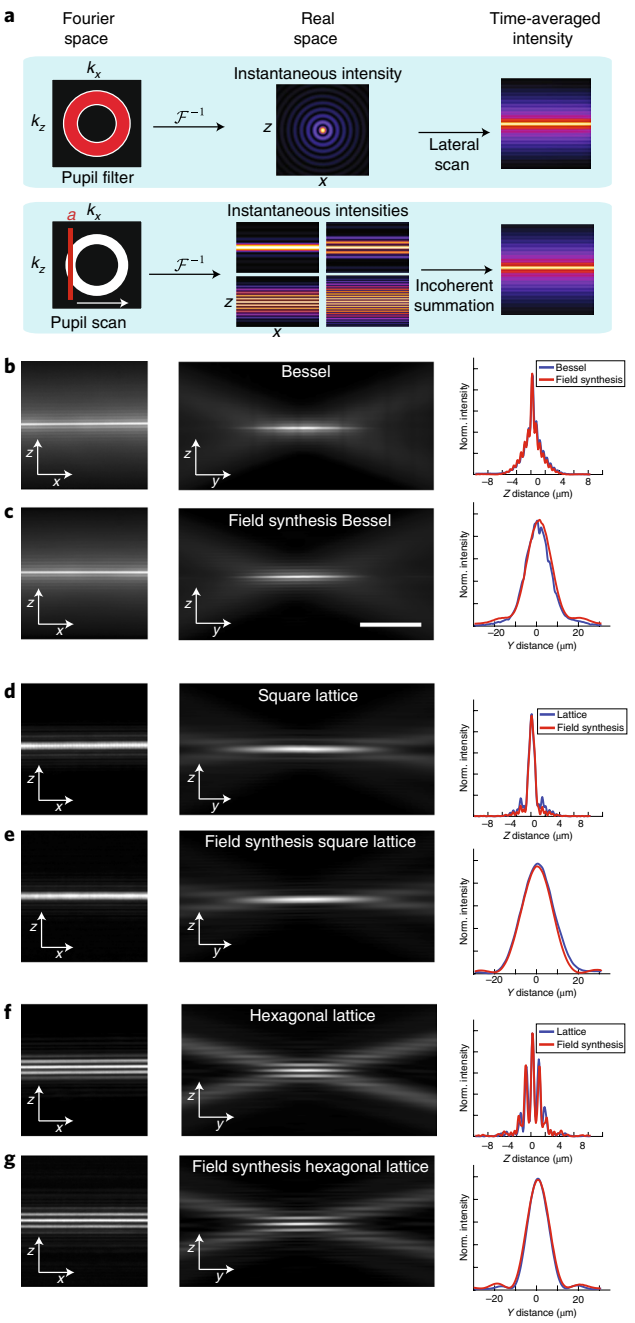

Fig. 1 | Light-sheet generation by field synthesis. a, Top, in light-sheet microscopy, a pupil filter conjugate to the back pupil plane of the illumination objective is used to shape the instantaneous laser focus at the front focal plane of the same objective. To generate a time-averaged sheet of light, the user rapidly scans the laser focus laterally. Bottom, the principle of field synthesis: a focused line is laterally scanned over a pupil filter, creating a time-averaged sheet of light in the front focal plane of the illumination objective. **b**,**d**,**f**, Experimental examples of a Bessel beam light sheet (**b**), square lattice (**d**) and hexagonal lattice (**f**) generated by traditional methods. **c**,**e**,**g**, Experimental examples of a Bessel beam light sheet (**c**), square lattice (**e**) and hexagonal lattice (**g**) generated with field synthesis. The cross-sectional profiles are averaged over six spatial regions in each image dataset. Norm., normalized. Scale bar, 10 μm (applies to **b**–**g**).

Unlike other attempts to simplify scanned Bessel beam light sheets or LLSM[8], field synthesis is expected to reproduce the time-averaged excitation intensity distribution of these methods exactly. Hence axial resolution, out-of-focus illumination and field of view can be tailored to the same degree as with the underlying original light-sheet techniques. Although the time-averaged intensity distributions obtained by field synthesis are predicted to be identical to dithered or scanned light sheets, there are important differences: (1) The spatial duty cycle (that is, the percentage of the field of view that is illuminated at a time) is 100% for symmetric pupil functions (as used in Bessel and LLSM) because the light sheet is synthesized from a series of cosines (which have maxima at their origin). As a consequence, field synthesis cannot be combined with virtual confocal line detection in scanned light sheets[9–11], as the entire field of view is illuminated instantaneously. (2) Field synthesis, as implemented here, is based on refraction and thus can be implemented with achromatic optics, permitting simultaneous multicolor excitation.

To evaluate this optical principle, we compared the three-dimensional (3D) intensity distributions of Bessel and lattice light sheets with those generated by field synthesis (see Methods for details). As seen in Fig. 1b–g, both in cross-sectional views and in the propagation direction, field synthesis produces identical light sheets. For LLSM, it has been hypothesized that complex 2D interference patterns are necessary for its advantageous properties, such as the aspect ratio of the sheet and suppression of its side lobes. We found that we could obtain the exact same light sheets by incoherently summing one-dimensional intensity patterns (Supplementary Notes 1 and 3, Supplementary Figs. 2–7 and Supplementary Software), even in the presence of non-ideal line scans (Supplementary Fig. 8).

To compare rates of photobleaching between a Bessel beam LSFM constructed using conventional beam scanning and field synthesis, we measured the fluorescence intensity of 100-nm fluorescent microspheres and microtubule plus-end tracking proteins (labeled with EB3-mNeonGreen) in osteosarcoma (U2OS) cells over time. In both cases, owing to its two-orders-of-magnitude-higher spatial duty cycle, field synthesis resulted in lower rates of photobleaching (Fig. 2a–c and Supplementary Video 3). The low spatial duty cycle of scanned Bessel beams has been previously identified as the cause of increased photobleaching and spurred the development of LLSM[5,12,13]. Our results show a new, general mechanism to improve spatial duty cycle for propagation-invariant beams without requiring complex 2D interference patterns and retaining the original light sheet[14–17].

Compared with LLSM, field synthesis improves the spatial duty cycle theoretically by a factor of two as estimated via numerical simulations of the intensity distribution in the main lobe of the light sheet. Practically, our measurements indicate that field synthesis matches the excellent low photobleaching of a dithered square lattice (Supplementary Fig. 9) and provides comparable biological imaging performance (Supplementary Fig. 10), while offering other advantages such as simultaneous multicolor imaging. To demonstrate that field synthesis can be used as an alternative to LLSM, we performed sensitive dual-color imaging of MV3 melanoma cells labeled with a PI3K biosensor (GFP-AktPH) and a membrane-localized fluorescent protein (tdTomato-membrane). Both conventional and field synthesis LLSM allowed us to resolve fine cellular features such as small membrane protrusions and filopodia and provided excellent optical sectioning. However, because field synthesis is compatible with simultaneous multicolor imaging, the volumetric imaging rate was twofold faster than conventional LLSM when two channels were acquired, despite the use of identical imaging parameters (camera integration time, z-step size and so on). This allowed us to observe rapid subcellular events such as small membrane protrusions and filopodia dynamics, which were temporally undersampled in the sequential acquisition mode afforded by conventional LLSM (Fig. 2d–k and Supplementary Videos 4–6). While sequential

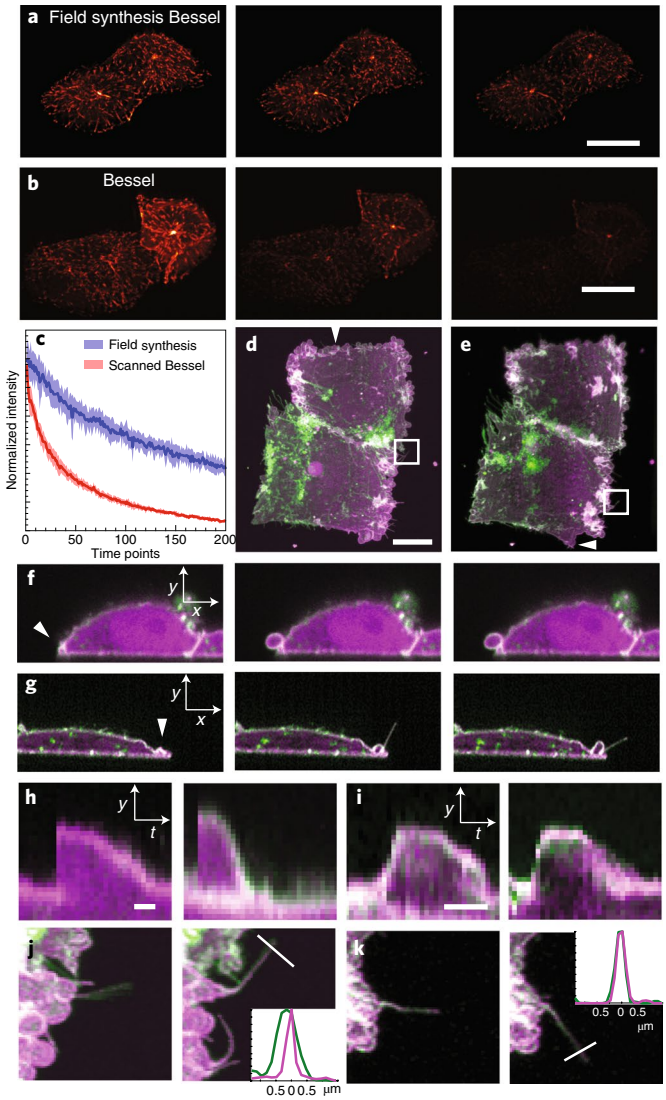

Fig. 2 | Field synthesis reduces photobleaching and allows for simultaneous multicolor imaging. a, Three time points (first, fifteenth and thirtieth, from left to right) of a movie of EB3 in a U2OS cell as imaged with field synthesis. **b**, Three time points (first, fifteenth and thirtieth, from left to right) of a movie of EB3 in a U2OS cell as imaged with scanned Bessel beam light-sheet microscopy. **c**, Comparison of photobleaching rates of 100-nm fluorescent microspheres over 200 acquired volumes as imaged with field synthesis and Bessel light-sheet microscopy. Shaded bands represent the 95th percentile of each photobleaching curve; the solid line represents the mean (11 beads were measured for each condition). **d**, MV3 cancer cell as imaged with sequential dual-color square LLSM at a volumetric rate of 0.23 Hz. Membrane labeled with tdTomato-membrane is shown in green, and the GFP-AktPH biosensor is shown in magenta. **e**, The same cell as in **d**, imaged with field synthesis in square lattice mode at a volumetric rate of 0.47 Hz. **f**, Three consecutive time points of a cross-section at the position indicated by an arrowhead in **d**. **g**, Three consecutive time points of a cross-section at the position indicated by an arrowhead in **e**. **h**, Kymographs of two blebs imaged with square LLSM. Onset of bleb formation is discontinuous. **i**, Kymographs of two blebs imaged with field synthesis. Bleb growth is gradual owing to the increased sampling frequency. **j**, Magnified view of the boxed region in **d** showing two time points of filopodia dynamics. **k**, Magnified view of the boxed region in **e** showing two time points of filopodia dynamics. Scale bars, 10 µm (**a**,**b**,**d**) or 20 s (**h**,**i**). Repeatability: in **a**,**b**, four cells were imaged in each light-sheet mode and a similar difference in photobleaching was observed. **d**–**k**, Seven cells were imaged in the lattice mode and nine cells in the field synthesis mode, yielding similar results.

LLSM acquisition could, in principle, be accelerated by the use of shorter camera integration times, this would come at the cost of a loss of signal or higher peak illumination intensities, and might not be possible owing to camera bandwidth limitations. Thus, we believe that simultaneous multicolor imaging with field synthesis expands the capabilities of LLSM, particularly in spectral imaging applications[18].

In conclusion, we have introduced with the field synthesis theorem a novel and universal mechanism to generate scanned light sheets, including dithered lattices. Field synthesis simplifies the optical train compared to LLSM. The consequence is that there are fewer critical components to align and temporally control, which eases the building and operation of such an instrument, both practically and financially. Additionally, simultaneous multicolor acquisition can be achieved. Field synthesis was not designed to allow structured illumination LLSM through a single objective[5], but it is conceptually compatible with structured illumination with multiple objectives[19]. In our view, such multi-objective illumination will allow users to unlock the full potential of structured illumination microscopy with light sheets, and we believe that field synthesis will be more experimentally tractable than a conventional lattice setup for such a configuration.

Beyond LLSM, the field synthesis principle applies to any type of intensity distribution. With a simple change of the pupil filter, other light-sheet types can be generated, including Bessel, Gaussian and Airy light sheets. Furthermore, our insight that any light sheet can be synthesized by a superposition of one-dimensional functions also implies that field synthesis could be used to rationally design new forms of light sheets, each uniquely optimized to specific merit functions. Thus, we believe that field synthesis will make LSFM modular and democratize advanced techniques, including but not limited to LLSM.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at https://doi.org/10.1038/s41592-019-0327-9.

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Author contributions
R.F., K.M.D. and B.-J.C. designed the research. B.-J.C. and R.F. designed, built and operated the microscope. R.F., K.M.D., P.R. and B.-J.C. performed image analysis. E.S.W. provided the MV3 cancer cells. M.K. carried out mathematical derivations with P.R. and R.F., proved the theorems and prepared the MATLAB code. R.F., K.M.D., M.K. and B.-J.C. wrote the manuscript, and all authors provided feedback.

Competing interests
R.F., B.-J.C. and M.K. have filed a patent for the field synthesis process and its applications to microscopy.

Additional information
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Methods

Microscope setup. The basic units to generate lattice and field synthesis light sheets are shown in Supplementary Fig. 1, and our microscope setup is illustrated in Supplementary Fig. 11. A 488-nm (Sapphire 488–300 LP, Coherent) and a 561-nm (OBIS 561-150 LS, Coherent) laser are combined with a dichroic mirror (LM01-503-25, Semrock), and focused through a 50-µm spatial filter (P25H, Thorlabs) with a 50-mm-focal-length achromatic lens (AC254-050-A, Thorlabs). Thereafter, flip mirror 1 is used to select either the LLSM or field synthesis optical paths, and the beam is recollimated with an achromatic lens with 75-mm (AC254-075-A, Thorlabs) or 400-mm (AC508-400-A, Thorlabs) focal length, respectively. Further, the optical system is operated in a fluorescence (for example, with orthogonal detection; Supplementary Fig. 11a) or transmission (for example, with diascopic detection; Supplementary Fig. 11b) mode, to image fluorescent specimens and investigate light-sheet properties, respectively.

The LLSM path is a simplified version of the original setup[5]. The collimated and expanded beam is further magnified and shaped to a line beam by a pair of achromatic cylindrical lenses (68-160, Edmund Optics, and ACY254-200-A, Thorlabs). The line beam uniformly illuminates a region on the spatial light modulator (SLM, SXGA-3DM, Forth Dimension Displays) where the lattice pattern is displayed. A polarized beam splitter (10FC16PB.3, Newport) and a half-wave plate (AHWP10M-600, Thorlabs) are placed in front of the SLM, and together with the SLM they form the pattern generator[5,20]. The lasers are arranged in such a way that the polarization states fit the pattern generator. The diffraction light from the SLM is focused by an achromatic lens (AC508-400-A, Thorlabs) on a custom-designed mask (mask 1, Photo Sciences, Inc.). The mask has various annuli to block unwanted diffraction orders (mainly the zeroth order, but also higher orders). Different sizes of the annuli correspond to different outer and inner numerical apertures (NA) on the back focal plane of the illumination objective lens. After passing through the mask, the desired diffraction orders are de-magnified through two achromatic lenses (AC254-150-A and AC254-100-A, Thorlabs) and fit into the size of a galvanometer (galvo mirror, 6215 H, Cambridge Technology). A folding mirror, an achromatic lens (AC-254-100-A, Thorlabs), flip mirror 2 and a tube lens (ITL-200, Thorlabs) then direct the light to the illumination objective lens (40×/0.8-NA, 3.5-mm working distance (WD); CFI Apo 40XW NIR, Nikon) to form a light sheet in sample space. All distances between a pair of lenses are equal to the sum of their focal lengths, so any pair of lenses forms a 4f arrangement. Thus, the SLM is conjugated to the focal plane of the illumination objective lens. The mask is at the Fourier plane of the SLM and is conjugated to the back focal plane of the objective lens. As the galvanometer is conjugated to the mask and the back focal plane of the objective lens, it scans the light sheet laterally (y) on the focal plane of the detection objective; that is, it enables rapid lateral scanning of a Gaussian or Bessel light sheet and the dithered mode of the lattice light sheet. Supplementary Fig. 7 shows typical lattice interference patterns that we could obtain with our setup.

The field synthesis path consists of an iris (IDA20, Thorlabs), an achromatic cylindrical lens (ACY254-050-A, Thorlabs), an achromatic lens (AC254-075-A, Thorlabs), another custom-designed mask (mask 2) and a tube lens (ITL-200, Thorlabs). An alternative beam expander design to illuminate efficiently smaller fields of view is discussed in Supplementary Note 4 and Supplementary Fig. 12. A galvanometer (galvo mirror, 6215 H, Cambridge Technology) and a folding mirror are placed between the achromatic cylindrical lens and the achromatic lens. The galvo mirror is conjugated to the sample plane, which is a similar configuration as used in mSPIM[21]. This path shares the same final tube lens and illumination objective lens with the lattice LSFM path. All distances between a pair of lenses are equal to the sum of their focal lengths, so any pair of lenses forms a 4f arrangement. The galvanometer in the field synthesis path is conjugated to the focal plane of the illumination objective lens. The mask is conjugated to the back focal plane of the objective lens, which is conjugated to the same plane as the mask in the lattice LSFM path. Because the magnification between the mask and the back focal plane of the objective lens is different in the field synthesis LSFM path and the lattice LSFM path, the sizes of the annuli on mask 2 are designed to match the outer and inner NAs resulting from the annuli on mask 1, as shown in Supplementary Table 2.

The detection path is orthogonal to the illumination path as in a conventional LSFM setup and consists of a detection objective lens (40×/0.8-NA, 3.5-mm WD; CFI Apo 40XW NIR, Nikon), a tube lens (ITL-200, Thorlabs), a dichroic mirror (Di03-R561-t3-25x36, Semrock) that separates the green and the red fluorescence and two identical sCMOS (scientific complementary metal-oxide semiconductor) cameras (Orca Flash4.0 v.2, Hamamatsu). A red fluorescence emission filter (FF01-593/LP-25, Semrock) and a green fluorescence emission filter (ET525/50m, Chroma) are placed in front of camera 1 and camera 2, respectively. A piezo stage (P621.1CD, Physik Instrumente) is used to move the sample diagonally to the detection axis in the fluorescence mode (Supplementary Fig. 11a) or move the detection objective lens in the transmission mode (Supplementary Fig. 11b).

In the transmission mode (Supplementary Fig. 11b), the detection objective lens is placed along the propagation of the light sheet, and an achromatic lens (AC508-400-A, Thorlabs) is used as the tube lens. An ND filter (ND40A, Thorlabs) is placed in front of the sCMOS camera (Orca Flash4.0 v.2, Hamamatsu) to reduce the intensity of the straight light from the lasers. The combination of the 400-mm tube lens and the 40× objective lens offers a final magnification of 80×, which results in one pixel on the camera corresponding to 81.25 nm and that can present the light sheet with a reasonable contrast and resolution. By stepping the detection objective lens with the piezo stage through the illumination lattice, we investigate and measure the 3D light-sheet properties.

The pattern on the back focal plane of the excitation objective lens can be observed by camera 4 (DCC1545M, Thorlabs) with an ND filter (ND10A or ND20A, Thorlabs) in the front when flip mirror 3 is used. Two achromatic lenses (AC254-100-A and AC254-050-A, Thorlabs) are used to image the back focal plane of the objective lens, which is conjugated to the plane of the masks, on the camera. Observing the pattern on this plane allows us to verify the final outer and inner NAs from both masks and adjust the scan range of the galvo mirror in field synthesis.

Microscope control and image acquisition. The data acquisition computer was equipped with an Intel Xeon E5-2687W v.3 processor operating at 3.1 GHz with 10 cores and 20 threads, 128 GB of 2,133 MHz DDR4 RAM and an integrated Intel AHCI chipset controlling 4 × 512 GB solid-state hard drives in a RAID0 configuration. All software was developed using a 64-bit version of LabView 2016 equipped with the LabView Run-Time Engine, Vision Development Module, Vision Run-Time Module and all appropriate device drivers, including NI-RIO drivers (National Instruments). Software communicated with the camera (Flash 4.0, Hamamatsu) via the DCAM-API for the Active Silicon Firebird frame-grabber and delivered a series of deterministic transistor–transistor logic triggers with a field programmable gate array (PCIe 7852R, National Instruments). These triggers included analog outputs for control of mirror galvanometers, piezoelectric actuators, laser modulation and blanking, camera fire and external trigger and triggering of the ferroelectric spatial light modulator. A timing diagram for the galvanometric beams scanning in field synthesis illustrated in Supplementary Fig. 13. All images were saved in the OME-TIFF format[22].

Some of the core functions and routines in the microscope control software are licensed under a material transfer agreement from Howard Hughes Medical Institute, Janelia Farm Research Campus. The control software code for field synthesis can be requested from the corresponding authors and will be distributed in accordance with this material transfer agreement.

Data analysis. Image analysis was carried out with Fiji[23] and MATLAB 2017a (Mathworks). To evaluate the rate of photobleaching for 100-nm fluorescent microspheres, we measured the average intensity of 11 similarly bright beads through time by recording their peak intensity at every time point. To this end, a MATLAB script was written that detected the peak intensity of all beads in the dataset, and a subset was manually selected that yielded the same average initial intensity for both imaging modalities. The same script was used to initially calibrate the power levels that yielded similar fluorescent counts when the same sample region was imaged with either Bessel beam or field synthesis light sheets. Both the bead imaging and the EB3 imaging in the U2OS cells were conducted with such calibrated power levels.

Deconvolution and data post-processing. The point-spread function (PSF) used for deconvolution of the lattice light-sheet (conventional and field synthesis) data was synthesized with the PSF Generator plugin in ImageJ[23,24]. The Born and Wolf 3D optical model was used to generate the PSF. Illumination and detection PSFs were synthesized in 3D with the practical experimental parameters (that is, the refractive index, the wavelength, the NA and the voxel size). The final light-sheet PSF is the product of the illumination and detection PSFs. Because our sample was placed diagonally to the illumination and detection, and we scanned the sample to acquire the images (sample-scan acquisition), the light-sheet PSF was sheared oppositely relative to the direction of the sample scan. Three-dimensional deconvolution was performed with ten iterations of the Richardson–Lucy algorithm as implemented in MATLAB. The deconvolved data were sheared at an angle of 45° using an affine transform and bleach corrected with Fiji using the histogram matching method.

For microtubule plus-end-tracking protein data, we obtained experimental PSFs by imaging isolated 100-nm nanospheres with both conventional Bessel and field synthesis modes of microscope operation. We chose experimental PSFs here, as the Bessel beam light sheet has a more complex structure and thus could not be approximated with the numerical PSF Generator. These data were deconvolved, the z axis bicubically interpolated so that the voxel dimensions were isotropic (162.5 nm) and sheared at an angle of 45° using an affine transform. The EB3 data were not bleach corrected.

Biological sample preparation. Both metastatic melanoma (MV3; a kind gift from P. Friedl, MD Anderson Cancer Center) and osteosarcoma (U2OS; a kind gift from R. McIntosh, University of Colorado, Boulder) cell lines were cultured in DMEM supplemented with 10% FBS and penicillin–streptomycin and maintained at 37 °C with 5% CO_2 atmosphere. MV3 cells were lentivirally transduced using the pLVX-IRES-PURO and pLVX-IRES-NEO expression systems. The PI3-kinase probe contains GFP fused to the PH domain of Akt, and the membrane marker contains tdTomato fused to the first 60 base pairs of neuromodulin. U2OS cells were

lentivirally transduced with a modified pLVX-shRNA2 vector with a truncated cytomegalovirus (CMV) promoter to reduce ectopic expression, as previously described[5]. SK-Mel2 cells (a kind gift from D. Drubin, University of California, Berkeley) stably expressing CLC-RFP from their native loci were engineered using zinc fingers and TALENS, cultured in DMEM/F12 supplemented with 10% FCS at 37 °C with 5% CO_2 atmosphere. All cells were plated on 5-mm coverslips and mounted in a custom sample holder for imaging.

Quantification of beam geometry. For direct measurements of the light-sheet intensity distribution, the excitation light was imaged in transmission with a 40× water-dipping objective, which was scanned in the *y* direction (for example, along the optical axis) with a 100-µm piezo actuator. With a 400-mm tube lens (Supplementary Fig. 11b), the effective pixel size of the imaged beam was 81.25 nm. Beam analysis software, written in MATLAB, identified the laser focus in *y*, the size of the beam waist in *z* and the beam propagation length in *y*. Because Bessel and lattice light sheets are propagation invariant (that is, they maintain their beam profile over extended propagation distances), we did not apply the Rayleigh length criterion to characterize the light sheets. Instead, we used the full width at half-maximum in the propagation direction.

Sequential lattice data. We developed a simple experiment, which can be reproduced by anyone who has access to an LLSM, to test the field synthesis theorem. We termed the procedure sequential lattice: here, we used the lattice light-sheet illumination path in the square lattice mode and manually positioned a mask to only transmit isolated diffraction orders along the *kz* direction (see also Supplementary Fig. 2). This was repeated for three mask positions, and the corresponding datasets were added numerically. As the summing of the intensity is incoherent, at no point does a 2D lattice pattern emerge; only one-dimensional intensity patterns are created. Compared to the dithered square lattice, the sequential lattice approach produces the same light sheets (Supplementary Fig. 2). In particular, some side-lobe structures are more closely reproduced than in Fig. 1d,e. We believe that these small differences arise from slightly different intensity distributions in the pupil plane. Overall, this further confirms that a dithered lattice light sheet can be synthesized by purely one-dimensional functions, as predicted by the field synthesis theorem.

Fluorescent measurement of light-sheet properties. We have characterized the light sheets using transmission measurements (Fig. 1 and Supplementary Fig. 11b), which provided images of the light sheets in the sample plane. Because of the involved imaging operation, blurring of the observed light sheet can be expected, but we argue that this effect will equally affect the conventional and field synthesis light sheets. Nevertheless, we also conducted a direct measurement of the light sheet in sample space. To this end, we mechanically scanned a 50-nm fluorescent bead in 0.2-µm steps in the *z*-direction and in 1-µm steps in the *y*-direction through a light sheet. At every scan position, a widefield image of the fluorescent emission from the bead was taken and the signal on the camera was summed up on a local window surrounding the bead. The window size was chosen so that it contained all diffraction rings of the bead at the largest defocus it was undergoing during the scan. The obtained intensity distributions of a square lattice and a field synthesis square lattice light sheet are shown in Supplementary Fig. 4. The width of the light sheet at its waist was 0.85 µm for the square lattice and 0.91 µm for the field synthesis light sheet. In the propagation direction, the full width at half-maximum was 16.7 for the square lattice light sheet and 17.2 µm for the field synthesis light sheet. These values are in close agreement with the transmission measurements shown in Supplementary Table 3, and the blurring in the transmission measurement appears to be small.

Statistics and reproducibility. For statistical analysis, we report the mean, s.d. and number of observations. To characterize the Bessel, lattice and field synthesis light sheets, each light sheet was spatially averaged across six different regions for multiple repetitions of the imaging experiment ($n = 2$–4; see Supplementary Table 3). In the main and Supplementary Figures, only the averaged light sheet (the mean) is shown, whereas in Supplementary Table 3, the mean and s.d. of the light-sheet parameters are provided. To further test reproducibility, each lattice and field synthesis light sheet was imaged eight times in transmission and was automatically processed with a MATLAB script. We found parameters for beam waist and length similar to the reported values in this paper. The fluorescence-based measurement of the lattice and field synthesis light sheets was also repeated eight times for each mode. We inspected the data visually and measured beam waist and length and found that the results were similar.

To compare the photobleaching of a digitally scanned Bessel beam light sheet and field synthesis, we evaluated the peak intensity of 11 individual beads. In Fig. 2c, the mean and 95th confidence interval for each time point is shown. Likewise, to compare photobleaching of a lattice light-sheet microscope and field synthesis, we evaluated the peak intensity of 11 isolated beads, and the mean and 95th confidence interval for each time point is shown in Supplementary Fig. 9. To compare the effect of photobleaching on EB3, we imaged four cells for each illumination mode. EB3 comets were automatically detected and their average intensity through time was measured. Both numerically and visually, the photobleaching was found to be similar to the results shown in Fig. 2a,b.

For the MV3 cells, at least seven cells were imaged in each mode, and similar bleb and filopodia dynamics were observed. On the particular MV3 cells that are shown in Fig. 2, we observed three filopodia blurring events in the sequential lattice mode and none in the simultaneous field synthesis mode.

For the resolution measurements on the clathrin-coated pits, we imaged five different cells for lattice and field synthesis. On the two cells we selected for visualization, we measured ten well-isolated pits for each imaging mode and computed the mean and the s.d. of the full width at half-maximum in the axial direction. On the other cells, we manually measured axial profiles with ImageJ and found similar values as reported in Supplementary Fig. 10.

Simulations. Simulations were written in MATLAB to verify the discrete mathematical proofs and their continuous analogs. MATLAB Live Scripts were created to integrate mathematical equations, demonstrative MATLAB code and corresponding illustrative figures while closely following the proofs of the field synthesis theorem. The MATLAB Live Scripts were published as HTML to facilitate viewing. The basic simulations were also ported to Python 2.7.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Code availability

The MATLAB code used in this work is available as Supplementary Software; any updated versions can be found at https://github.com/AdvancedImagingUTSW/FieldSynthesis. The instrument control software can be requested for academic use from the corresponding authors and will be delivered under material transfer agreements with HHMI and UT Southwestern Medical Center.

Data availability

The datasets acquired for this study are available from the corresponding author upon request.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Labview control software that was developed by Coleman technologies using Labview 2016
Data analysis	ImageJ, version 1.52i, Matlab versions 2016a-2018b

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No calculations to predetermine sample sizes were performed. In our experience, the intensity distributions of a light-sheet, which were characterized in this manuscript, are deterministic and do not vary much from experiment to experiment. We thus decided that a small number (eight) of measurements of the light-sheets would be sufficient to show repeatability. For cellular imaging, between four to seven cells were imaged for each imaging mode. This sample size was sufficient in our opinion since we were investigating a physical phenomenon (i.e. generation of light-sheets) and not a biological question. For the measurements of light-sheet parameters as shown in supporting table 3, the mean and standard deviation is calculated from over six positions across each light-sheet dataset and there are two to four light-sheet data sets, which leads to total sample size of 12-24 that was analyzed.The plots shown in fig. 1b-g, fig. s4, and fig. s6 are single line profile of each light sheet. To measure the axial resolution in a biological sample, 10 clathrin coated pits were measured for each imaging mode. As the resolution is governed by robust physical principles, we settled for this small sample size.
Data exclusions	No data were excluded intentionally.
Replication	We replicated our optical measurements of the various light-sheets eight time and the measurements were repeatable. We repeated the cell measurements from four to seven times depending on the experiment, and the results were reproducible.
Randomization	For the EB3 imaging in U2OS cell, we choose the cell randomly. For the imaging of MV3 cell, we choose the cell with strong fluorescence signal and desired morphology with nice blebs and filopodia. The SK-MEL2 cells were selected for having minimal background and having well isolated Clathrin coated pits.
Blinding	In principle, we were blinded to the group allocation since the cell was chosen randomly or by the morphology and the outcome is not related to cell morphology.

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Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
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☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	The MV3 cells were a gift from Peter Friedl, MD Anderson Cancer Center, Houston. The U2OS cells were a gift from Dick McIntosh, University of Colorado, Boulder. SK-MEL2 cells were a gift by David Drubin, University of California, Berkeley.
Authentication	Both the MV3, SK-MEL2 and U20S cells were not authenticated.
Mycoplasma contamination	All cells were tested and did not show mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	We checked the ICLAC data base and none of the cells that we have used is listed in it. To our best knowledge, these ar not commonly misidentified cell lines.